Results. As shown in Figure 4, NextLat consistently outperforms all baselines in Countdown. Notably, even with a shallow supervision horizon of $d = 1$, NextLat substantially surpasses MTP and JTP trained with the same horizon (**>35.7% improvement**). To better understand this gap, we analyzed the equations generated by each model and evaluated their equation validity, i.e., whether the computed left-hand side equals the right-hand side. As shown in Figure 5, most calculation errors occur in the final equation (Eq. 3). This indicates that the model's lack of planning capability results in its realization of being unable to achieve the goal only at the end. Unable to revise earlier missteps, it forces an invalid final equation to match the desired outcome—a behavior termed *the regretful compromise* by Ye et al. [2025]. NextLat demonstrates **stronger lookahead planning**: even with $d = 1$, it achieves substantially higher mean validity in the final equation (54.8%) compared to the next best baseline (42.3%). This suggests that the latent-state prediction objective may help the model anticipate long-range dependencies and form globally consistent plans, reducing the tendency to make myopic errors. NextLat also achieves leading performance with horizons $d = 4$ and 8.

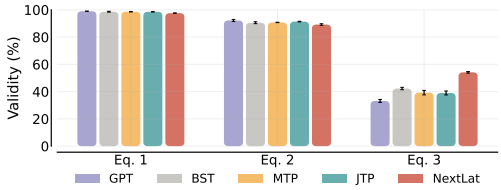


Figure 5: Validity of equations (i.e., LHS = RHS) generated on Countdown. All models in this plot use $d = 1$.

4.3 Planning

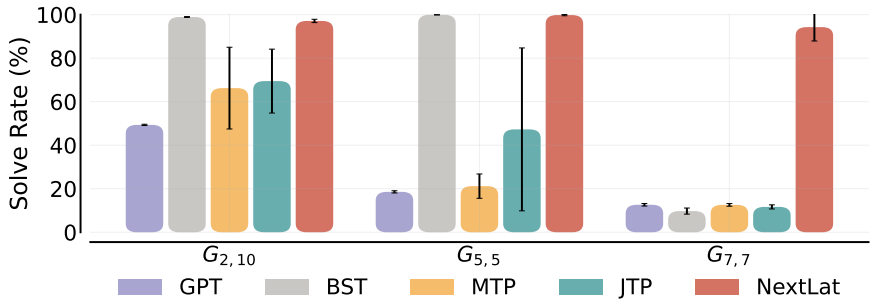


Figure 6: Accuracy on Path-Star graph task. Unlike the baselines, NextLat maintains close to 100% solve rate for all graphs.

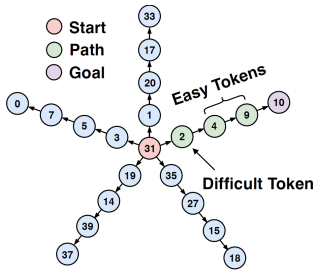


Figure 7: Illustration of a $G_{5,5}$ Path-Star graph [Bachmann and Nagarajan, 2024].

A Path-Star graph [Bachmann and Nagarajan, 2024] $G_{d,\ell}$ consists of a center node and d disjoint arms, each consisting of $\ell - 1$ nodes. Figure 7 depicts an instantiation of the $G_{5,5}$ topology. A training instance is a tokenized sequence that contains the edge list, the start and end nodes, and the correct path from start to end. This task represents a minimal instance of lookahead planning, a core capability underlying more complex behaviors such as storytelling. Yet, despite its apparent simplicity, next-token prediction models struggle to solve it.

Setup. Following Bachmann and Nagarajan [2024], we generate 200k training samples and set $N = 100$, such that node values in each graph are randomly drawn from $1, \ldots, 100$. For MTP, JTP, and NextLat, we set the multi-step prediction horizon to $d = \ell - 2$, ensuring that the target (end) node lies within the multi-step prediction horizon of the center node. We evaluate performance across three graph configurations: $G_{2,10}$, $G_{5,5}$, and $G_{7,7}$ across five random seeds per baseline.

Results. As shown in Figure 6, NextLat maintains close to 100% solve rate for all topologies of the Path-Star graphs. BST, while able to solve $G_{2,10}$ and $G_{5,5}$, begins to fail at the larger graph $G_{7,7}$. Note that our results differ from that presented in Hu et al. [2025] (BST) and Ahn et al. [2025] (JTP). In their setup, they use a much smaller problem settings of $N = 50$, and generate a fresh batch of graphs every iteration. On the other hand, we use the original (and more difficult) setup which has a fixed sample size of 200k and $N = 100$.

The Path-Star graph task is specifically designed to reveal the myopic behavior of teacher-forced next-token prediction models, which tend to exploit local shortcuts instead of learning to perform lookahead planning necessary to solve the task. This phenomenon, termed the *Clever Hans cheat* [Bachmann and Nagarajan, 2024], is related to the difficulty of learning parity [Hu et al., 2025] and has motivated methods such as BST and JTP that attempt to mitigate shortcut learning through multi-token predictions. However, these approaches operate in token space, making them still susceptible to local n-gram regularities that do not capture the underlying transition structure required for long-horizon planning. In contrast, NextLat performs prediction in latent space, enforcing recurrent transition consistency at the representation level. NextLat's success across all graph configurations, unlike MTP, JTP, and BST, suggests that latent-space prediction better avoids shortcut learning and yields more generalizable solutions. Furthermore, given the limited training samples (i.e., 200k) in this task, data efficiency is crucial. NextLat's strong performance in this low-data regime suggests that it extracts more useful supervision from each training sequence than token-prediction baselines, consistent with our analysis in Section 3.1.

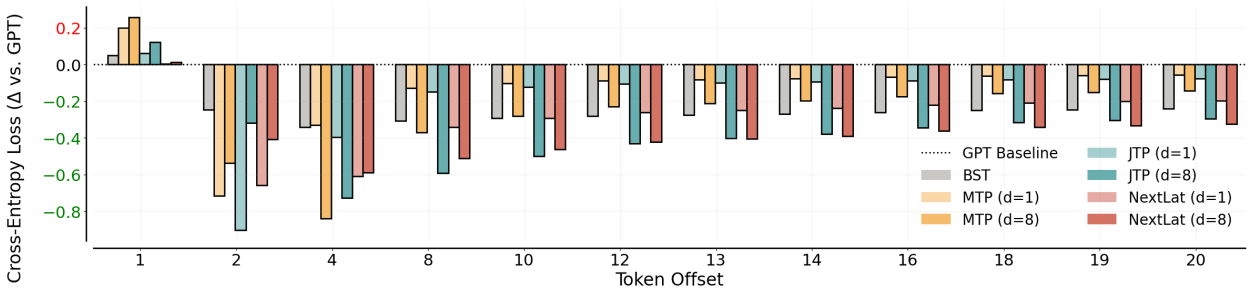


Figure 8: Cross-entropy loss difference relative to GPT, obtained from linear probes trained on frozen hidden states to predict tokens at varying token offsets (x-axis values) ahead. Lower values indicate better predictive performance.

Next, we compare the models on TinyStories [Eldan and Li, 2023], a dataset consisting of synthetic short stories. Storytelling is inherently a long-horizon planning problem; a coherent narrative requires maintaining persistent entities, tracking causal relationships, resolving conflicts and delayed resolutions, and satisfying narrative constraints across many timesteps. Generating such sequences therefore depends not only on next-token prediction, but on belief-state–like abstractions that encode information predictive of future story trajectories.

Setup. Following Hu et al. [2025], we tokenize the dataset of 2.7 million stories into a vocabulary of 1,000 tokens and construct training sequences of length 256. All models are trained for 100k steps, which is sufficient for convergence. We include comparisons of transformers trained using MTP, JTP, and NextLat with multi-step prediction horizons $d \in \{1, 8\}$. After training, we freeze the model parameters and train 20 independent linear probes, one per token offset, to predict tokens at offsets $1, \ldots, 20$ steps ahead from the hidden states of the frozen models using the same dataset. This allows us to assess whether the models' representations encode information predictive of future tokens, or just local token correlations.

Results. We plot the difference in probe performance relative to probes trained on GPT's hidden states in Figure 8. For clarity, we display only selected token offsets here (see Figure 16 in the appendix for full results). Observe that the additional token-level prediction objectives in BST, MTP, and JTP consistently cause significant degradation in next-token prediction (i.e., token offset = 1). Moreover, probe performance on JTP and MTP representations declines sharply with increasing token offset. This indicates that these multi-token prediction models, lacking guarantees of learning belief states, could encode information useful only for short-horizon prediction. In contrast, NextLat matches GPT's next-token performance across both $d \in \{1, 8\}$ and exhibits the strongest long-range predictive capability (up to 20 tokens ahead) for both $d = 1$ and $d = 8$. These results suggest that NextLat's latent-state objective induces belief-like representations that encode predictive information about future events—an ability essential for maintaining coherence in long-range narrative generation tasks like TinyStories.

4.4 Language Modeling

Model	FW-Edu ppl ↓	Wiki. ppl ↓	LAMB. ppl ↓	LAMB. acc ↑	PIQA acc ↑	HellaS. acc ↑	Wino. acc ↑	ARC-e acc ↑	ARC-c acc ↑	SIQA acc ↑	SciQ acc ↑	Avg.
GPT	**10.52**	**17.93**	20.26	42.07	73.45	**58.79**	60.46	68.18	39.16	42.32	86.10	58.82
JTP (d=1)	11.08	19.28	21.88	41.35	**74.92**	57.43	58.64	68.73	39.25	42.99	87.30	58.83
JTP (d=2)	11.18	19.60	22.11	41.37	73.34	56.84	59.98	68.86	38.57	**43.35**	86.70	58.63
MTP (d=1)	10.90	18.82	20.23	41.26	74.32	58.05	**60.54**	68.52	38.91	42.84	85.40	58.76
MTP (d=2)	11.00	18.61	18.34	43.43	72.80	57.92	59.35	68.35	39.08	41.97	86.60	58.69
NextLat (d=1)	10.83	18.39	19.77	41.08	73.07	58.35	59.27	69.65	39.68	43.24	86.00	58.79
NextLat (d=2)	10.88	18.44	**17.83**	**43.86**	73.61	57.79	59.20	**69.74**	**40.10**	41.91	**87.50**	**59.21**

Table 2: Downstream language modeling evaluation on 1.3B-parameter models trained on 100B FineWeb-Edu tokens. Best scores are in bold and second-best are underlined.

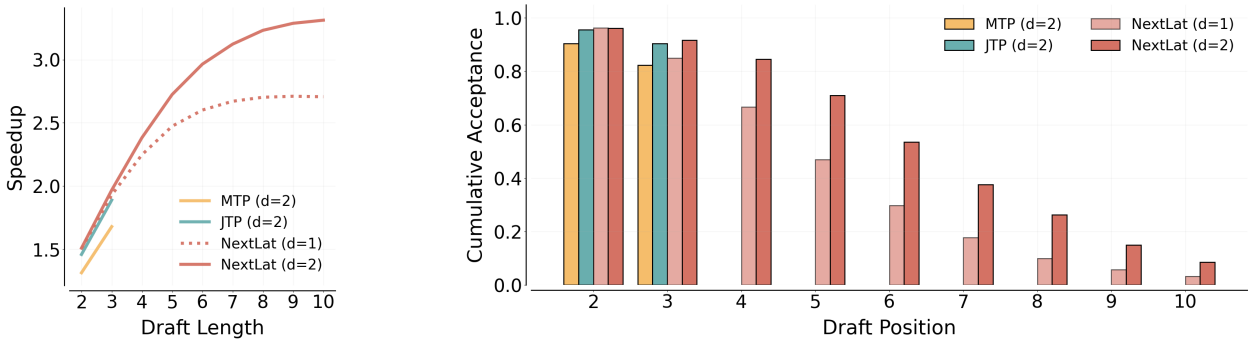
Setup. We pretrain 1.3B-parameter models on 100B tokens from the FineWeb-Edu dataset [Penedo et al., 2024], excluding BST due to its high computational cost. After pretraining, we use the LM Evaluation Harness [Gao et al., 2024] to evaluate the zero-shot accuracy of the models on multiple-choice language modeling benchmarks. We also evaluate the self-speculative decoding performance of the multi-step prediction models, i.e., JTP, MTP, and NextLat, across Wikipedia, Books, Code, and Math domains. For each dataset, we sample 1024 prompts of length 512 tokens and generate 512-token continuations using the speculative sampling algorithm of Leviathan et al. [2022]. For each model, we report the average number of accepted tokens per drafting step, as well as the inference speedup relative to standard autoregressive sampling from the base transformer, measured on $8\times$NVIDIA B200 GPUs. For multi-step prediction models, we primarily focus our analysis on training with horizons of $d = 1$ and $d = 2$. MTP and JTP operate in token space and are therefore usually limited to speculative decoding within the fixed multi-token horizon used during training. Our aim is to highlight the advantages of NextLat's variable-length self-speculative decoding, which enables drafting beyond the training horizon. To this end, we vary NextLat's speculative draft length between 2 and 10 tokens and report the highest inference speedup achieved for each domain.

	Wikipedia		Books		Code		Math	
Model	Speedup	Accepted Tokens	Speedup	Accepted Tokens	Speedup	Accepted Tokens	Speedup	Accepted Tokens
JTP (d=1)	1.46	0.96	1.47	0.97	1.47	0.98	1.46	0.97
JTP (d=2)	1.88	1.84	1.90	1.89	1.88	1.85	1.89	1.86
MTP (d=1)	1.38	0.91	1.39	0.95	1.40	0.97	1.39	0.95
MTP (d=2)	1.68	1.72	1.72	1.83	1.75	1.91	1.72	1.84
NextLat (d=1)	2.68	3.52	2.72	3.64	2.29	2.66	2.30	2.72
NextLat (d=2)	**3.21**	**4.59**	**3.32**	**4.86**	**2.38**	**2.83**	**2.87**	**3.94**

Table 3: Relative speedup and average accepted tokens per drafting steps over diverse domains. Note that "Accepted Tokens" excludes the next-token prediction which is always accepted.

Results. In Table 2, we present the language modeling perplexity (ppl) and zero-shot accuracy (acc) of the pretrained models across several benchmarks. Consistent with observations in Gloeckle et al. [2024], we do not observe significant improvements in multiple-choice task accuracy over standard next-token training (GPT) when using multi-token prediction objectives. NextLat ($d = 2$) does show a modest gain in average accuracy over GPT (59.21 vs. 58.82), but these improvements are not consistent across tasks. Larger model sizes might be necessary to see more significant improvements. Notably, NextLat **better preserves next-token perplexity compared to MTP and JTP** across FineWebEdu, Wikitext, and LAMBADA. This is consistent with our earlier observations on TinyStories in Section 4.3. Preserving high fidelity in next-token prediction is arguably important, as prior work shows that lower pretraining perplexity correlates with improved aggregate downstream and post-fine-tuning performance at larger model scales [Gadre et al., 2024, Zhang et al., 2026].

Next, in Table 3, we show the self-speculative decoding results. Observe that the average accepted tokens per drafting steps for NextLat far exceeds its training horizon d, indicating that the learned latent dynamics remains coherent over extended rollouts. This further highlights the strong long-range predictive capability of the induced belief state representations. Crucially, this ability to support variable-length self-speculative decoding enables NextLat to achieve **substantially higher inference speedups than MTP and JTP across all domains**. Figure 9 further shows speedup and cumulative acceptance rate versus draft length on the FineWeb-Edu validation set. Speedup increases sublinearly with draft length, reaching up to $3.3\times$, and fully valid (i.e., all tokens accepted) drafts persist even at length 10. This demonstrates a clear benefit to drafting tokens beyond the training horizon with NextLat. In Section F.6.1, we extend our comparisons to include JTP and MTP trained with larger horizons ($d = 4$). Note that the training cost of JTP and MTP increases substantially with larger d (see Table 4), making JTP ($d = 4$) and MTP ($d = 4$) significantly more expensive to train than NextLat ($d = 1, 2$). Still, even with longer multi-token supervision, these baselines still fail to surpass NextLat in speculative decoding performance. This highlights a key advantage of NextLat's variable-length speculative decoding: it enables long speculative drafts while requiring training only at shallow, computationally efficient horizons.



(a) (b)

Figure 9: (a) Inference speedup ratio of JTP, MTP, and NextLat on the FineWeb-Edu validation set. (b) Cumulative acceptance rate of draft tokens (product of per-position acceptance probabilities), indicating how likely the entire speculative draft remains valid (i.e., all tokens accepted) as draft length increases.

5 Discussions

5.1 Comparison Against Current Approaches

	GPT	BST	MTP	JTP	NextLat
Training parameters ($d = 1, 2, 8$)	1.32B	2.57B	1.37B / 1.42B / 1.72B	1.34B	1.40B
Inference parameters	1.32B	1.32B / 2.57B	1.32B	1.34B	1.32B
Training steps/second ($d = 1, 2, 8$)	3.09	0.89	2.80 / 2.58 / 1.70	3.15 / 2.92 / 2.02	3.09 / 2.79 / 1.73
Gradients	$O(T)$	$O(T^2)$	$O(Td)$	$O(Td)$	$O(Td)$

Table 4: Comparison of training speed, parameter count and gradient signals provided on FineWeb-Edu pretraining. For training parameter numbers and training iterations/second, we report values for prediction horizons $d = 1, 2, 8$. If only a single value is shown, it means that the values are the same for all horizons. All training speeds are measured on a single NVIDIA B200 GPU with batch size of 33k tokens.